Lecture 14:
Regression discontinuity I

PPHA 34600
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From last time: wrapping up DD(D)

We met the DDD estimator:

$$
\begin{aligned}
Y_{ijt} = \beta_0 &+ \beta_1 Treat_i + \beta_2 Post_t + \beta_3 Affected_j + \beta_4 (Treat_i \times Post_t) \\
&+ \beta_5 (Post_t \times Affected_j) + \beta_6 (Treat_i \times Affected_j) \\
&+ \tau (Treat_i \times Post_t \times Affected_j) + \varepsilon_{ijt}
\end{aligned}
$$

A tour of research designs

We've met several research designs this quarter:

1. **Randomized controlled trial**
   - Eliminates selection bias via randomization
2. **Regression adjustment & matching**
   - Selection on observables
   - Strong assumption: We observe everything that might matter
3. **Instrumental variables**
   - Use some (quasi)-random variation to move endogenous treatment
   - Strong assumption: Exclusion restriction
4. **Panel fixed effects**
   - Compare units to themselves over time
   - Strong assumption: Parallel counterfactual trends

My favorite selection on unobservables design

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- **Basic intuition:** Use a (policy-induced) cutoff to compare i and j

$\rightarrow$ Look at "barely-treated" units vs. "barely-untreated" units

- Enables us to come close to mimicking random assignment
- Identifying assumptions are transparent
- We can do a lot of this in pictures

Regression discontinuity design

As usual, we'd like to run:

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

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The regression discontinuity:

- Suppose D_i is determined by whether or not X_i lies above a cutoff, c
- We call X_i the "running variable" here
- **Idea**: Having X_i just above or just below c is as good as random...
- ... And there is a discontinuous change in D_i as a result of crossing c

→ We can compare Y_i for units with X_i just above c to Y_i for units with X_i just below c

Sharp regression discontinuity

In the most straightforward, or “sharp” RD design:

- $Pr(D_i = 1|X_i \geq c) = 1$ and $Pr(D_i = 1|X_i < c) = 0$
- $Pr(D_i = 1|X_i \geq c) - Pr(D_i = 1|X_i < c) = 1$
- Nobody with $X_i < c$ gets treated
- Everybody with $X_i \geq c$ gets treated
- The probability of treatment jumps from 0 to 100% as X_i crosses c
- $D_i = \mathbf{1}(X_i \geq c)$

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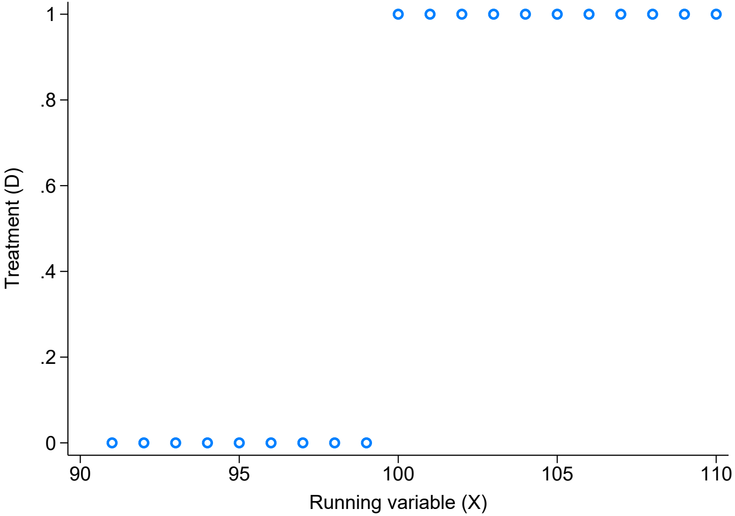
- $D_i = \mathbf{1}(X_i \geq c)$

$\rightarrow$ This is equivalent to **perfect compliance** in the RCT

Sharp regression discontinuity: Treatment assignment



Sharp regression discontinuity: Estimation

To get τ, compare units with $D_i = 0$ and $D_i = 1$ exactly at the cutoff:

$$\hat{\tau}^{SRD} = E[Y_i(1) - Y_i(0) | X_i = c]$$

- The estimator is defined exactly at the cutoff
- But we will never observe $Y_i(D_i = 1, X_i < c)$ or $Y_i(D_i = 0, X_i \geq c)$

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To overcome this, get super close to c but not exactly there:

$$\underbrace{\lim_{x \downarrow c} E[Y_i | X_i = x]}_{\text{approach } c \text{ from above}} - \underbrace{\lim_{x \uparrow c} E[Y_i | X_i = x]}_{\text{approach } c \text{ from below}}$$

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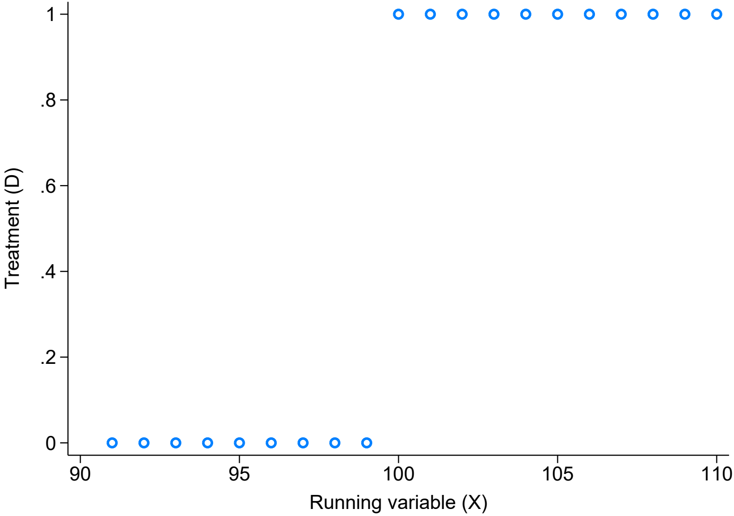
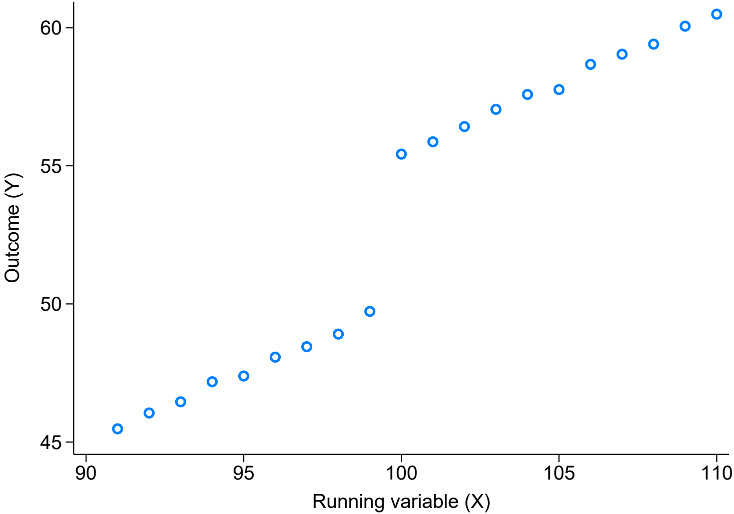
$$= \underbrace{\lim_{x \downarrow c} E[Y_i(1)|X_i = x]}_{X_i \geq c \text{ get treated}} - \underbrace{\lim_{x \uparrow c} E[Y_i(0)|X_i = x]}_{X_i \geq c \text{ don't get treated}}$$

Sharp regression discontinuity: Treatment assignment



Sharp regression discontinuity: Outcomes



Sharp regression discontinuity: Identifying assumption

We only need one identifying assumption for the sharp RD:

$$E[Y_i(1)|X_i = x] \text{ and } E[Y_i(0)|X_i = x] \text{ are continuous in } x$$

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In words:

- We can compare units with X_i very close to, but not exactly at, c

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In words:

- We can compare units with X_i very close to, but not exactly at, c

In other words:

- The cutoff is as good as randomly assigned

In more other words:

- There are no discrete jumps in Y_i at c except due to D_i

In even more other words:

- All observed and unobserved determinants of Y_i (other than treatment) are smooth around the cutoff

RD validity tests

We can perform two major RD validity checks:

1. A "bunching" or "manipulation" test
2. A "covariate smoothness" test

$\rightarrow$ As usual, we can't prove the identifying assumption!

$\rightarrow$ We can just provide evidence in favor of it!

Manipulation tests

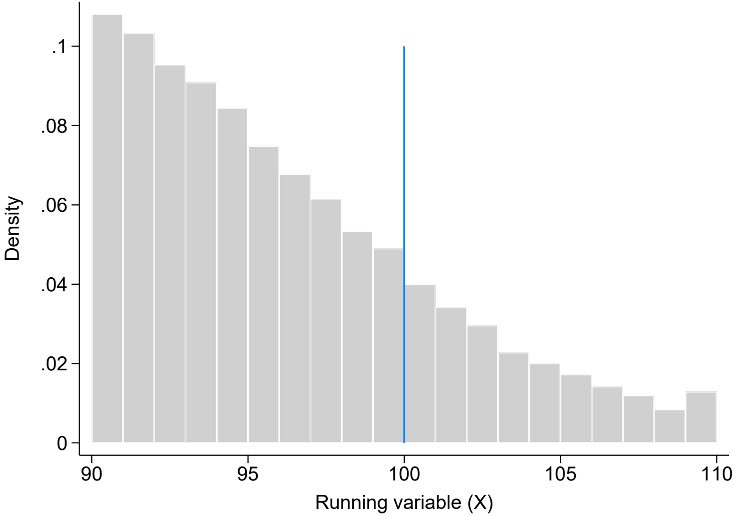
We are assuming that $X_i - c$ is as good as randomly assigned:

- (In the neighborhood of c)

$\rightarrow$ We want to make sure units can't sort around c

$\rightarrow$ We test this by looking at the distribution of X_i

A manipulation test



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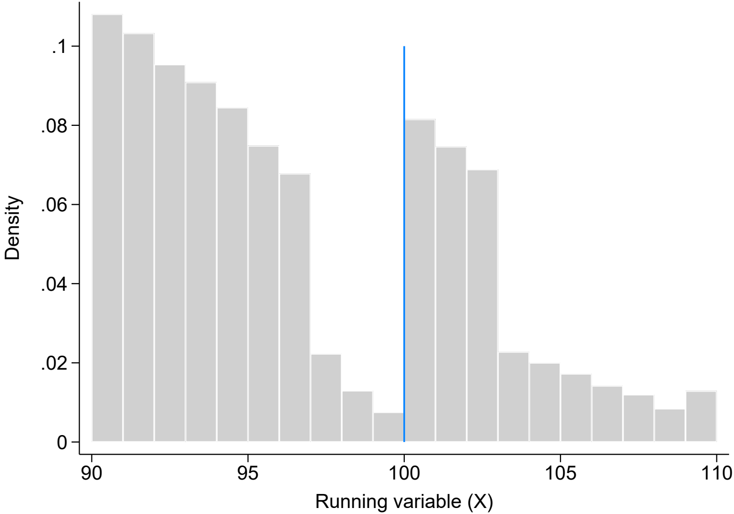

Covariate smoothness

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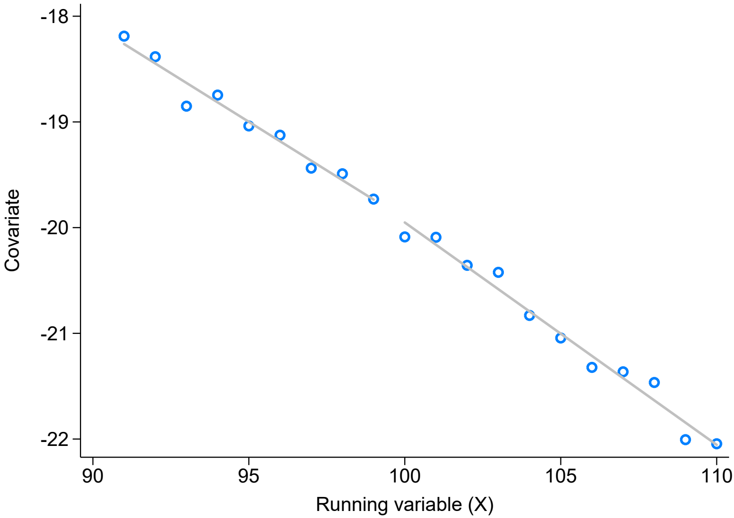
- This works for predetermeined variables only!
  - $\rightarrow$ Outcomes affected by treatment may well jump at c
- This test is imperfect (why?)
  - $\rightarrow$ We can't check for smoothness of unobservables

A covariate smoothness test



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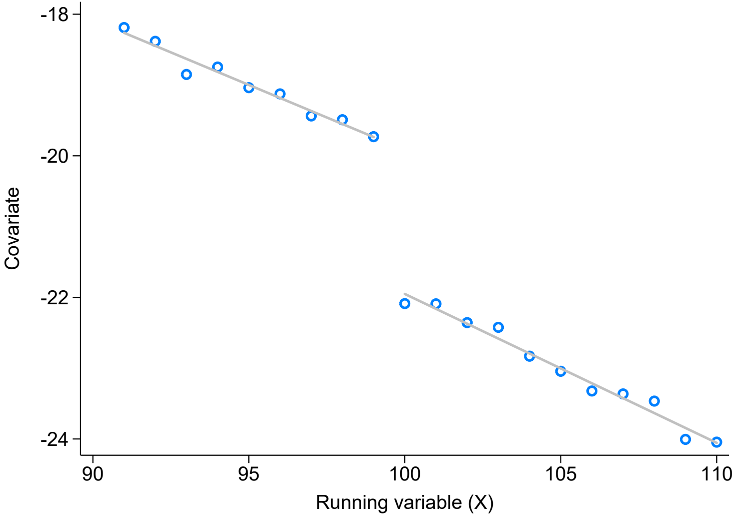

Putting the “regression” in regression discontinuity

We want the difference in outcomes for just-treated vs. just-untreated:

$$\tau^{SRD} = E[Y_i(1) - Y_i(0)|X_i = c] = \lim_{x \downarrow c} E[Y_i|X_i = x] - \lim_{x \uparrow c} E[Y_i|X_i = x]$$

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We estimate average outcomes just below and above the cutoff:

$$\hat{\tau}^{SRD} = \bar{Y}(D_i = 1; c \leq X_i \leq c + h) - \bar{Y}(D_i = 0; c - h \leq X_i < c)$$

where $c - h \leq X_i \leq c + h$ is the bandwith in which we're "close" to c

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This leads to the regression-based RD:

$$Y_i = \alpha + \tau D_i + \varepsilon_i \text{ for } c - h \leq X_i \leq c + h$$

where $D_i = \mathbf{1}[X_i \geq c]$

How do we choose h?

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- We want h to be small: the RD is identified only at c
    - If too small, we will get imprecision (no sample density)
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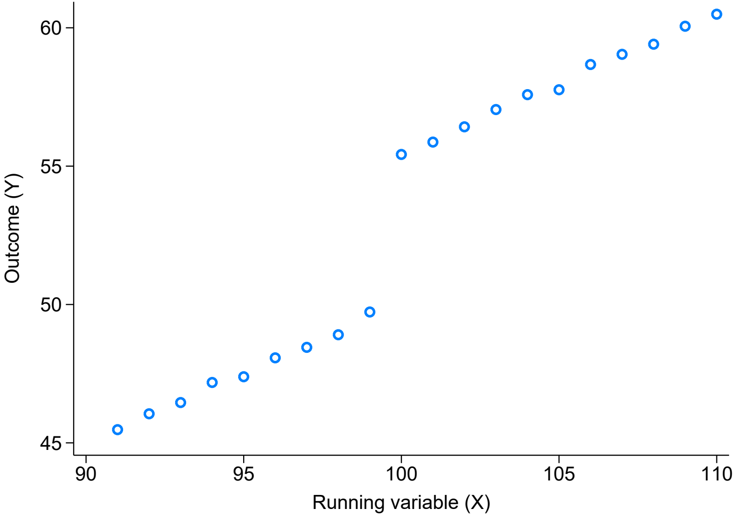
→ This is another example of the **bias-variance tradeoff**

Bandwidth-induced bias in the RD



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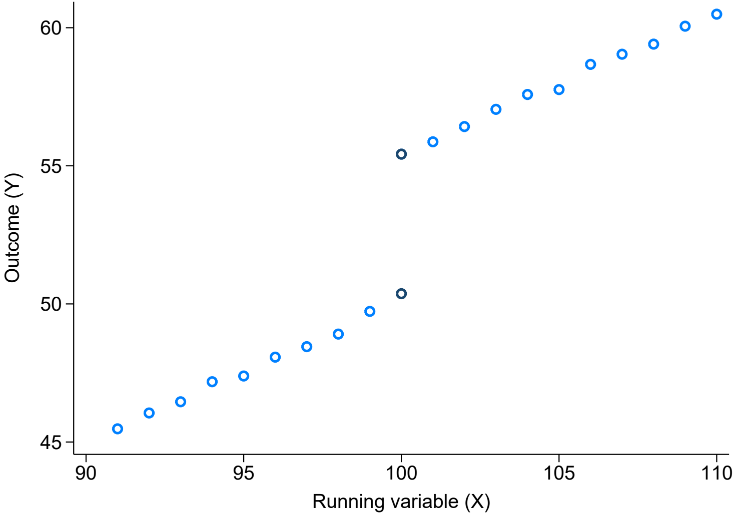

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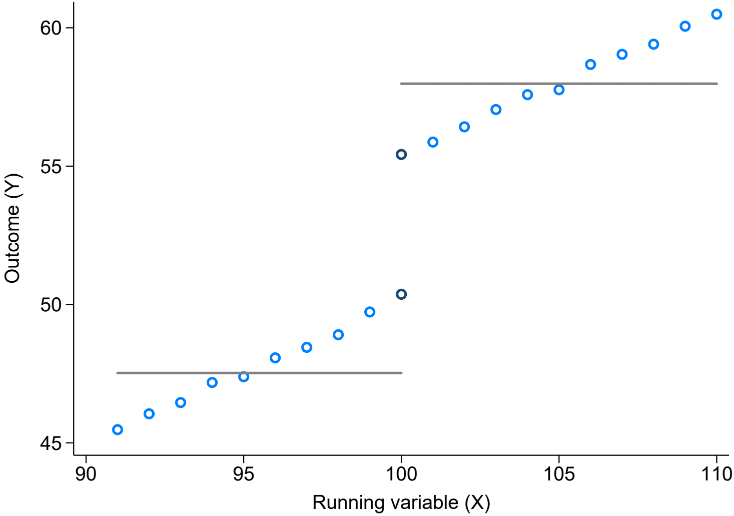

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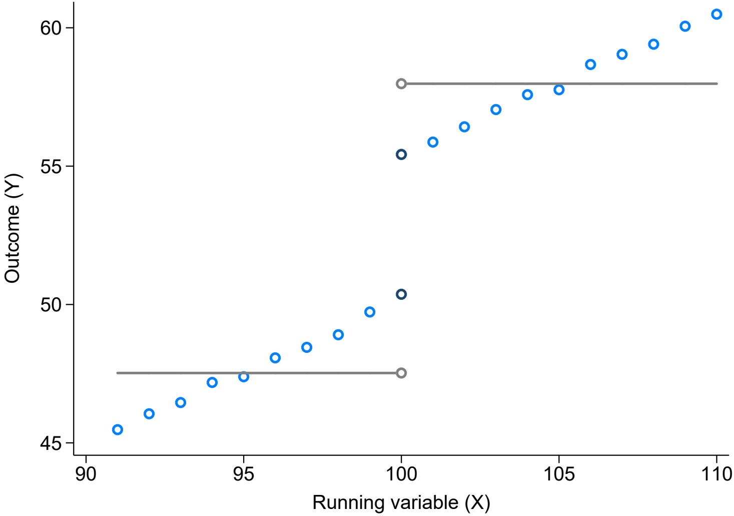

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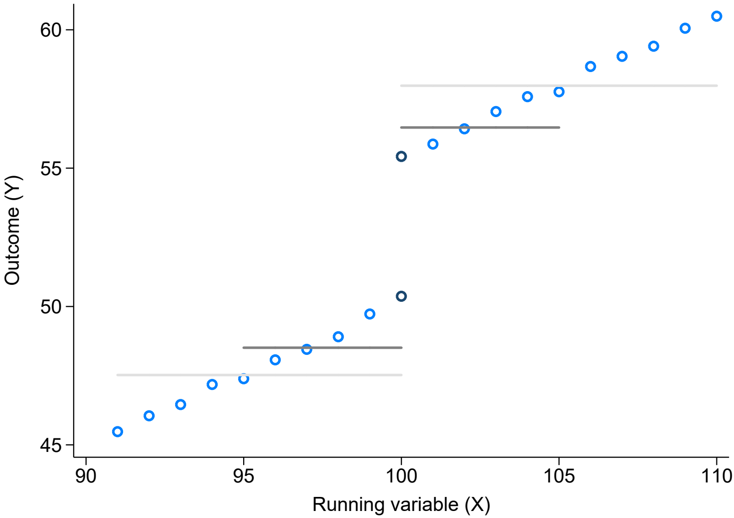

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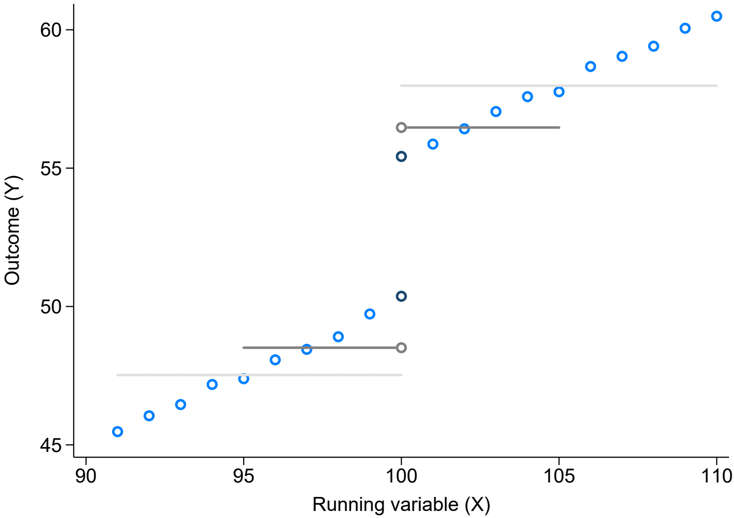

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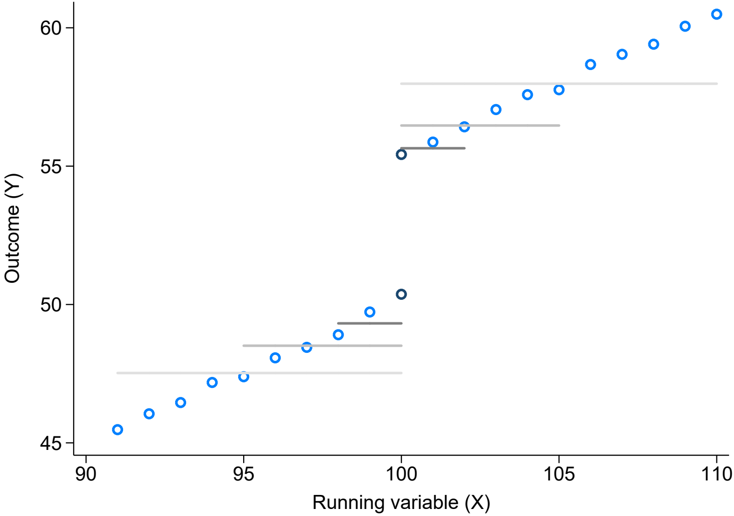

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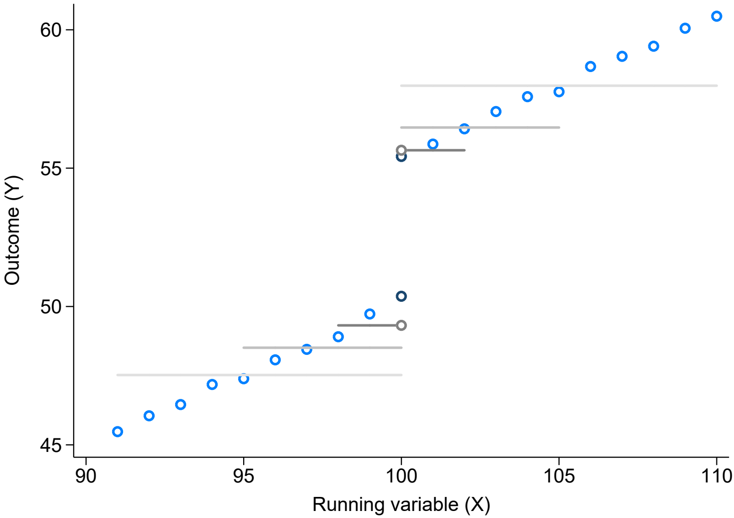


Can we do better than differences in means?

Simple differences in means ignores any relationship between Y_i and X_i:

- We can improve on this by controlling for the underlying relationship
- If we know $Y_i(X_i)$ is a linear function, we can just regress:

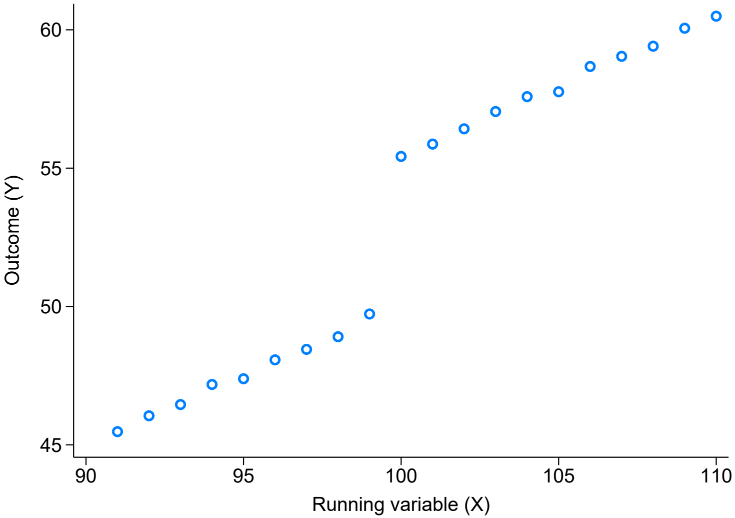
$$Y_i = \alpha + \tau D_i + \beta(X_i - c) + \varepsilon_i$$

$$= \alpha + \tau \mathbf{1}[X_i \geq c] + \beta(X_i - c) + \varepsilon_i$$

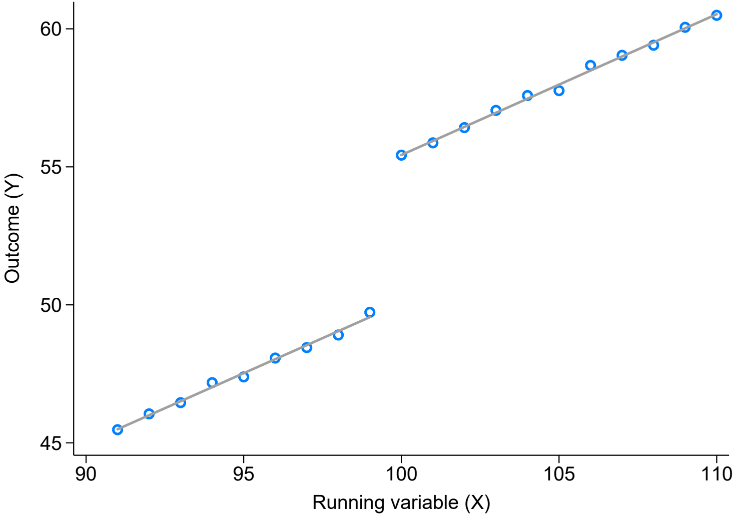
- We can also allow for different slopes above and below c:

$$Y_i = \alpha + \tau D_i + \underbrace{\beta_1(X_i - c)}_{\text{slope below}} + \underbrace{\beta_2(X_i - c)D_i}_{\text{slope above}} + \varepsilon_i$$

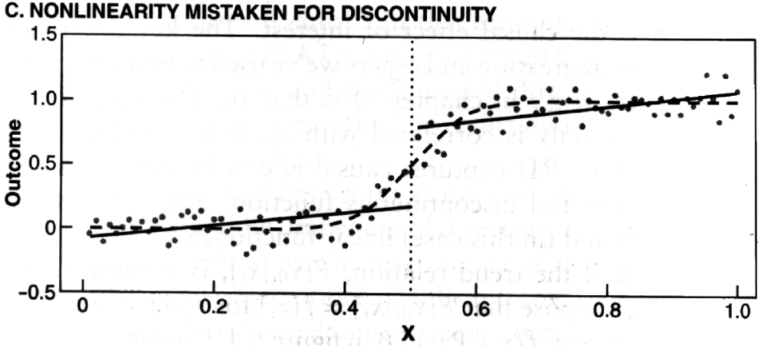
Controlling for the slope



Controlling for the slope



We have to be a bit careful about this!



A note on interpretation

External validity is an important consideration for RD:

- (This is true for all designs...)
- ... but in RD, we are estimating results *at the cutoff, c*
- In sharp RD, we're estimating a LATE around the cutoff!

→ This may be different (or not) from the ATE, or other LATEs from other cutoffs

An example: Electricity rebates in California

Policy issue:

- Economists love taxes (to reduce bad stuff)!
- And electricity generation produces bad stuff
- ...but nobody else likes taxes, so policymakers often use rebates instead
- Are these rebates actually effective?

Approach:

- Look at the "20/20" policy in California
- Customers who reduced energy use by 20% received a 20% discount
- Eligibility for the policy isn't random...
- ...but is determined by a policy rule:
- Customers had to have an account before a cutoff date

→ Use a RD model to estimate treatment effects

Estimating the effects of 20/20

Koichiro runs a version of:

$$Y_{it} = \tau D_{it} + f(X_i) + \alpha_i + \delta_t + \varepsilon_{it} \text{ for } c - h \leq X_i \leq c + h$$

where

Y_{it}: energy use by household i in month t

$D_{it} = \mathbf{1}[X_i \leq c] \times [t \in \text{program period}]$

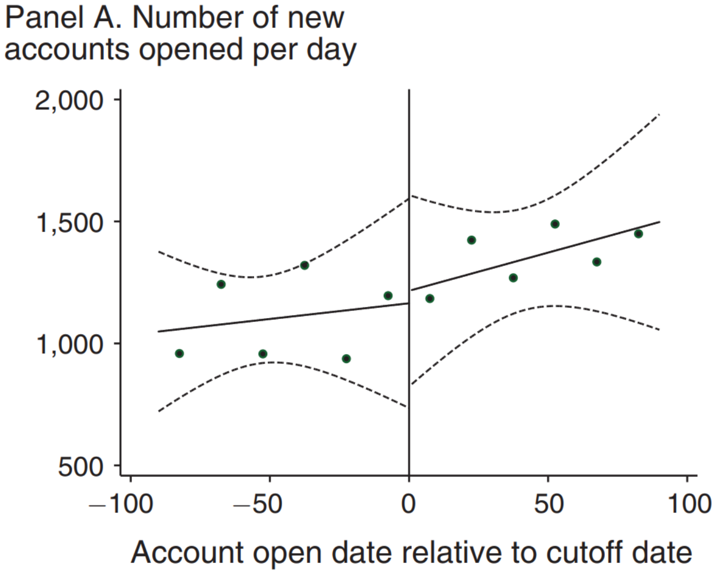
c is a cutoff date by which accounts had to be opened for eligiblity

$f(X_i)$ is a flexible function of the running variable, X_i

α_i, δ_t are customer and time FE, respectively

ε_{it} is an error term

Checking the identifying assumption



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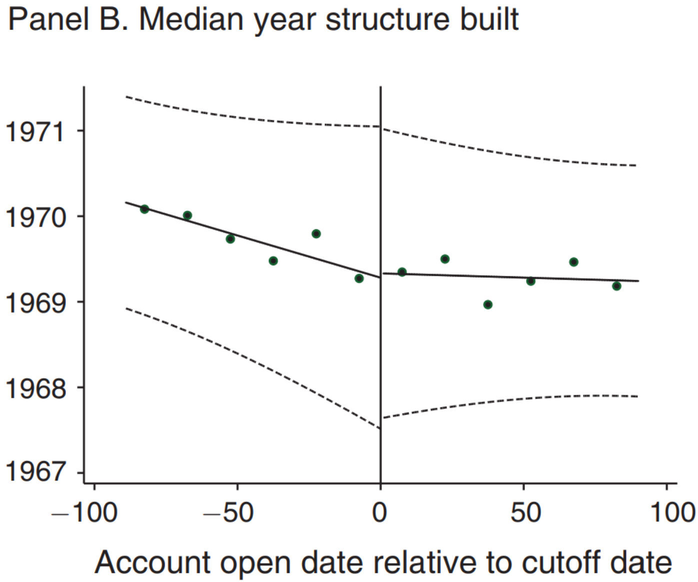

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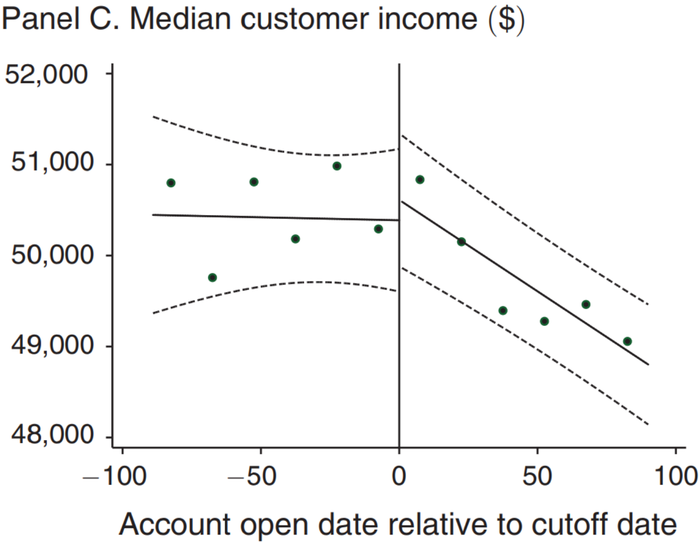

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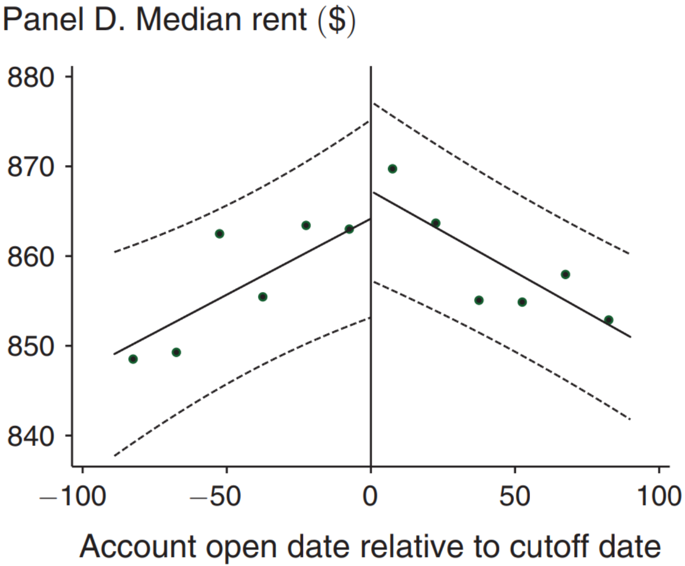
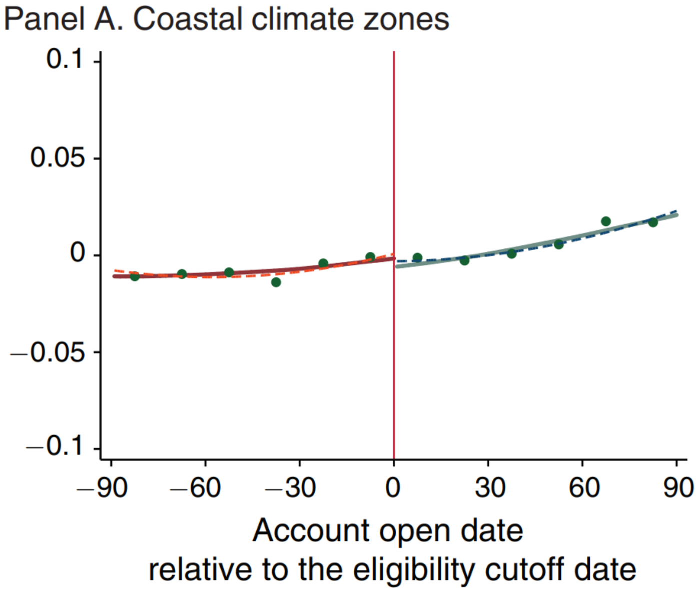


Main results



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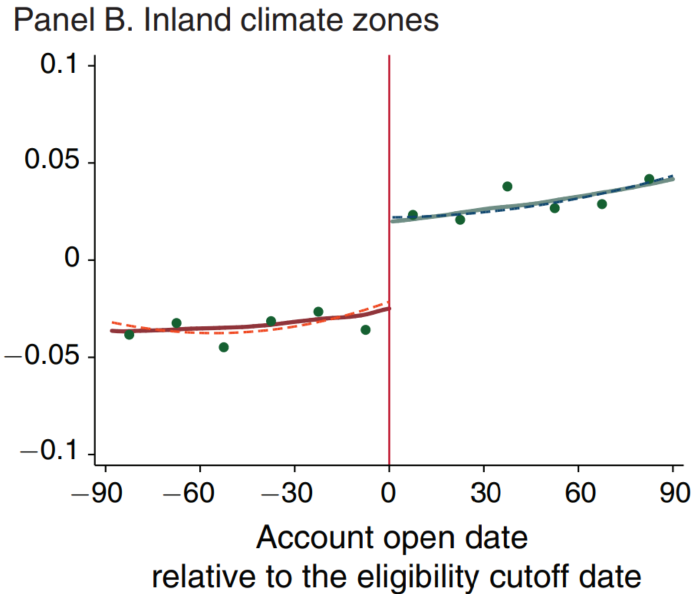

Bandwidth sensitivities

	Coastal climate zones			Inland climate zones		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment effect in May	0.004	0.003	0.005	−0.034	−0.039	−0.029
	(0.004)	(0.003)	(0.004)	(0.015)	(0.014)	(0.017)
Treatment effect in June	−0.002	−0.001	−0.003	−0.055	−0.059	−0.05
	(0.004)	(0.004)	(0.004)	(0.017)	(0.016)	(0.019)
Treatment Effect in July	0.004	0.005	0.005	−0.041	−0.039	−0.042
	(0.004)	(0.004)	(0.005)	(0.019)	(0.017)	(0.022)
Treatment effect in August	−0.004	−0.005	−0.003	−0.036	−0.034	−0.035
	(0.004)	(0.004)	(0.004)	(0.018)	(0.016)	(0.020)
Treatment effect in September	−0.005	−0.003	−0.004	−0.056	−0.053	−0.052
	(0.003)	(0.004)	(0.004)	(0.016)	(0.015)	(0.018)
Controls for $f(x)$	Local linear	Quadratic	Quadratic	Local linear	Quadratic	Quadratic
Bandwidth	90 days	120 days	60 days	90 days	120 days	60 days
Observations	2,540,472	3,325,388	1,707,589	208,537	237,264	162,067

Notes: This table shows RD estimates with different bandwidth choices and alternative controls for the running variable. The dependent variable is the log of electricity consumption. The standard errors are clustered at the customer level to adjust for serial correlation.

Cost-effectiveness

TABLE 8—PROGRAM COST PER ESTIMATED REDUCTIONS IN CONSUMPTION AND CARBON DIOXIDE

	Coastal	Inland	Total
Number of customers	3,190,027	299,178	3,489,205
Consumption in summer 2005 (kWh)	8,247,457,920	1,154,292,248	9,401,750,168
Direct program cost for rebate ($)	9,358,919	1,250,621	10,609,540
Estimated reduction (kWh)	9,908,840	50,605,714	60,514,555
Estimated reduction in carbon dioxide (ton)	4,459	22,773	27,232
Program cost per kWh ($/kWh)	0.945	0.025	0.175
Program cost per carbon dioxide ($/ton)	2,099	55	390
Program cost per carbon dioxide ($/ton) (Adjusted for noncarbon external benefits)	2,090	46	381

Recap

TL;DR:

1. The regression discontinuity is great
2. We can mimic an RCT in observational data
3. And the tests are visual and transparent

For next class

Topics:

- Regression discontinuity II

Reading: Chen, Ebenstein, Greenstone, and Li (2013). Be sure to also read:

- The supplementary material!